

SYKTYVKAR, Russian Federation

WMO: 238040

Lat: 61.6772N

Lon: 50.7847E

Elev: 119

StdP: 99.91

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.0	(c) -29.2	(d) -36.0	(e) 0.1	(f) -32.9	(g) -32.2	(h) 0.2	(i) -29.2	(j) 8.4	(k) -8.4	(l) 7.6	(m) -8.2	(n) 1.5	(o) 10	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.6	(c) 28.9	(d) 20.0	(e) 26.8	(f) 19.2	(g) 24.8	(h) 17.8	(i) 21.2	(j) 27.0	(k) 20.0	(l) 25.1	(m) 18.7	(n) 23.2	(o) 2.6	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.0	23.9	18.0	13.1	22.6	17.0	12.3	21.4	62.0	26.9	57.8	25.2	53.7	23.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.3	5.5	-36.1	31.3	3.4	2.0	-38.5	32.7	-40.5	33.8	-42.4	34.9	-44.9	36.4
			-35.9	22.8	3.5	1.5	-38.5	23.9	-40.5	24.8	-42.5	25.6	-45.0	26.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-13.9	-12.5	-5.5	2.0	9.0	14.4	17.4	13.9	8.7	2.0	-5.9	-10.9
	DBStd	12.22	8.20	8.08	5.40	5.01	5.43	4.62	4.13	4.17	3.91	4.29	6.69	8.45
	HDD10.0	3649	742	631	481	242	86	11	2	10	70	249	477	649
	HDD18.3	6163	1000	864	739	489	291	134	69	147	290	506	727	908
	CDD10.0	597	0	0	0	3	56	143	230	132	30	2	0	0
	CDD18.3	69	0	0	0	0	2	17	39	11	0	0	0	0
	CDH23.3	714	0	0	0	1	51	179	377	104	2	0	0	0
	CDH26.7	177	0	0	0	0	10	42	98	27	0	0	0	0
Wind	WSAvg	2.6	2.7	2.7	2.9	2.6	2.8	2.6	2.2	2.3	2.4	2.8	2.8	2.8
Precipitation	PrecAvg	569	36	29	28	31	44	68	67	67	55	58	49	39
	PrecMax	739	76	74	78	87	103	150	140	164	95	112	90	72
	PrecMin	396	11	4	5	0	11	11	13	8	19	14	10	9
	PrecStd	88	13	17	16	18	23	34	34	32	19	24	18	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	2.1	8.0	20.2	28.1	30.2	31.2	30.0	22.2	15.7	7.2	3.1
		MCWB	0.6	0.9	3.1	12.4	16.6	20.1	21.4	21.4	16.2	11.7	6.6	2.6
	2%	DB	0.0	0.8	5.0	15.2	24.2	27.8	29.2	26.1	19.0	11.8	4.2	1.2
		MCWB	-0.3	0.1	1.5	9.0	14.8	19.4	20.5	19.4	14.4	9.4	3.7	0.8
	5%	DB	-1.4	-0.2	3.0	11.9	21.7	25.2	27.2	23.5	16.8	9.8	2.7	0.4
		MCWB	-1.8	-0.7	0.7	6.6	13.3	17.6	19.7	17.7	13.1	8.3	2.2	0.0
	10%	DB	-3.3	-1.9	1.8	9.2	19.0	23.0	25.2	21.0	14.8	8.0	1.3	-0.7
		MCWB	-3.7	-2.6	0.1	5.0	11.9	16.2	18.6	16.3	12.2	7.0	0.8	-1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	1.1	3.9	12.9	18.4	21.7	22.8	22.0	17.0	12.1	6.3	2.5
		MCDB	0.7	1.6	6.3	19.8	24.8	27.7	28.7	28.4	21.1	14.1	6.8	2.8
	2%	WB	-0.4	0.1	2.2	9.5	16.0	20.3	21.6	20.1	15.2	9.9	3.6	0.9
		MCDB	-0.1	0.6	3.9	14.3	22.3	26.0	27.4	25.3	17.9	11.3	4.0	1.0
	5%	WB	-1.9	-1.0	1.1	7.2	14.4	18.9	20.5	18.5	13.7	8.2	2.1	0.0
		MCDB	-1.6	-0.4	2.5	10.8	20.4	23.8	25.9	22.1	16.1	9.4	2.4	0.3
	10%	WB	-3.8	-2.6	0.2	5.4	12.7	17.4	19.4	17.2	12.4	6.8	0.9	-1.1
		MCDB	-3.4	-2.0	1.4	8.5	17.7	21.7	24.2	20.2	14.6	7.8	1.2	-0.7
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.6	9.1	9.3	10.8	10.5	10.6	9.2	7.7	4.9	4.6	5.8
		MCDBR	6.1	5.3	9.7	14.1	15.3	13.9	13.7	13.0	11.3	7.7	4.0	4.0
		MCWBR	6.2	5.1	7.2	8.7	8.1	7.1	6.6	6.8	7.0	5.8	4.0	4.0
	5% WB	MCDBR	6.2	5.4	7.3	12.5	13.9	12.1	12.6	11.8	10.0	6.8	3.8	4.0
		MCWBR	6.3	5.3	5.7	8.3	8.1	6.7	6.5	6.5	6.6	5.6	3.8	4.1
Clear-Sky Solar Irradiance	taub		0.256	0.298	0.336	0.366	0.361	0.361	0.398	0.380	0.344	0.319	0.268	0.232
	taud		1.984	2.049	2.043	2.153	2.415	2.452	2.354	2.383	2.462	2.352	2.086	1.935
	Ebn at Noon		407	637	751	814	860	865	817	801	768	638	397	253
	Edh at Noon		41	76	110	121	102	100	108	97	75	58	36	24
All-Sky Solar Radiation	RadAvg		0.27	0.96	2.21	3.58	4.85	5.17	5.27	3.60	2.02	0.80	0.25	0.12
	RadStd		0.04	0.22	0.36	0.46	0.43	0.45	0.47	0.37	0.29	0.09	0.06	0.02

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page